Ernest Skrzypczyk

Bachelor of Science in Electrical Engineering - Control Engineering and Robotics

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Education

2015–current **Double Masters Degree**, *European Masters in Advanced Robotics + (EMARO+)*, University of Genoa, Italy / Ecolé Centrale de Nantes, France, A programme of Erasmus Mundus.

2007–2011 **Bachelor of Science - Electrical Engineer - Control Engineering and Robotics**, *Wrocław University of Technology*, Wybrzee Wyspiaskiego 27, PL-50370 Wrocław, *very good / bardzo dobry*.
ISCED 5A, Polish grading system; See supplement to diploma, page 5

2004–2006 **Abitur**, *Berufsbildende Schulen Meppen - gew. und kaufm. Fachrichtungen - Fachrichtung Technik*, Nagelhof 83, D-49716 Meppen,
3,0 - good / gut.
ISCED 3A, German grading system

Engineering project

Title *Metrological properties of a hybrid amplifier*

Supervisor PhD / dr in. Daniel Dusza

Description Design, construction and measurements of a hybrid amplifier with a working frequency within the audible range and a microcontroller based control system.

Grade excellent / celujący

Prizes and Awards Stowarzyszenie Elektryków Polskich - Association of Polish Electrical Engineers
1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011

Experience

Vocational

12.07.2010 – 20.08.2010 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein BerSSen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems. Internship for studies at the Wrocław University of Technology.

14.07.2008 – 15.08.2008 **Electrician** helper, *Elektro Westendorf GmbH*, D-49777 Klein BerSSen, Germany.
Electrical installations on construction sites. Also photovoltaic, other renewable energy and light systems.

2 weeks in May 2005 **Computer scientist**, *Ebert Mikrokomputer GmbH*, D-49716 Meppen, Germany.
Computer system installations mostly in the private sector. Customer care on site. Secondary school internship.

EMARO+

UNIGE **Projects and assignments:**

- *Localization of a biped humanoid robot using EKF within ROS framework (**Github repository**)*
- *Study and implementation of developed SLAM algorithms for a biped humanoid robot within ROS framework (**Github repository**)*
- *Mobile robots control: point and posture tracking, Lyapunov control, design and simulation*
- *Mobile robots motion planning using bug, distance based and probabilistic roadmap planning algorithms*
- *Offline and online tuning of EKF in a real world case scenario using landmarks*
- *Technical report on Agriculture in controlled environments*
- *Technical paper review and presentation on FLEXA*
- *Delmia simulations*
- *Design and evaluation of a user interface for a motion analysing application*
- *Basic image processing including edge and object detection, colour based segmentation, optical flow (Lucas-Kanade algorithm) and stereo vision*
- *POSIX compliant publisher / subscriber communication models using pipes and pthreads*
- *Realtime scheduling with RM, PS and EDF*

Languages

Polish	**Native**	
English	**B2+**	*Mark: very good / bardzo dobry at SJO WUT*
German	**C1**	*100% exam test score, 2nd place in contest at FLD WrUT*
Spanish	**A2**	*High school secondary language for 2 years*
Italian	**B1**	*Score 25/30, spent 1 year in Genova, Italy within EMARO+ programme*
French	**Pending**	*About to spent 1 year in Nantes, France within EMARO+ programme*

Computer skills

Operating systems: Extensive experience with Microsoft Windows, GNU/Linux, some experience with hardware specific solutions and FreeBSD

Distributions: Arch Linux (current main OS), OpenSUSE, Funtoo, Gentoo, Linux Mint, Ubuntu, Rasbian

Software experience	Various CAD/CAE, office and scientific applications and suites across all media fields including audio, video, graphics, text, design, simulation and engineering	Applications	LibreOffice, Microsoft Office, Altium Designer, Simatic Step 7, CX-Programmer, Delmia, Matlab, Spyder, Pythonxy, Scilab, Proto, Photoshop, Gimp, Inkscape, Ableton Live, Audacity, Geany, TeXstudio, nano
Shells / Command line interfaces	Zsh (current main shell), Bash, PowerShell, Command Prompt	Software and community contributions	Proto, Arch Linux Wiki articles (ALSA - Basic Syntax), Arch Linux Forums and packages maintenance, wine, q4wine

Programming languages

General languages and environments

- *Assembler on various hardware*
- *BASIC on Commodore 64 and various BASIC derivatives on robots*
- *C/C++ in courses, assignments, projects and contributions*
- *Python*
- *ROS - Robot Operating System*

PLC languages

- *Function block diagram*
- *Ladder logic*

Scripting

- *Bash / Zsh*
- *Matlab / Scilab / Pythonxy / Spyder*
- *Command Prompt / Batch*
- *Autohotkey*

Certificates, Licences

Electrical qualification	**Operation position** in the area of: **operation**, **maintenance**, **repairs** and **assembly**	*SEP - AoPEE, Wrocław* *Date of exam: 19.05.2010* *Valid for 5 years*
Driving licence	**Class A, B**	*Driving licence acquired in Germany in 2005*

Organisations, associations and memberships

SNS	Academic student association	*Took part in several projects, educational trip and made a publication for the SSC 2009*
STRIMER	Wrocław University of Technology	

Additional courses

Warsaw

2015	ABC of enterneurship
2015	Self presentation and interview skills

References

Title	Name	Position	Workplace
PhD / dr in.	Daniel Dusza ✉	assistant professor	Wrocław University of Technology, Institute of Electrical Machines, Drives and Measurements
Prof PhD / prof. dr hab. in.	Zdzisław Nawrocki ✉	professor, retired	Wrocław University of Technology, Institute of Electrical Machines, Drives and Measurements

Publications

Rzepecki Bartłomiej, Skrzypczyk Ernest, "*Plasma and its applications*", 7th Students' Science Conference, Wrocław, Poland, 2009

Prizes and Awards

Prize	1st place for best thesis in Electrical Engineering Department on Wrocław University of Technology in the academic year 2010/2011
Prize	Second place in German language tournament at FLD WrUT
Commendation	For best paper presentation at SSC 2009
Aknowledgment	Third place for best group work at SSC 2009

Scholarships

Several scholarships for achievements in studies at the WrUT Faculty of EE

Interests

Arts
- *Graphics, photography, image rendering and manipulation*
- *Music, audio creation and manipulation*
- *Paper model building*

Philosophy
- *New socio-economic structures*
- *Consequences of implementation of new technologies*

Technology
- *Self-sustainable autonomous vehicles, facilities and infrastructure*
- *Robot interaction*
- *Artificial intelligence*
- *FOSS, especially GNU/Linux*
- *Open Hardware, Open Technology, Open Economy*

Additional information

Mobility	Full mobility within 157 countries, especially across Europe
Self-development	Main focus was to shift working environment to and towards FOSS, contribute and exchange with the FOSS community.